# MILITARY SPECIFICATION

## INTERCONNECTING GROUP, ON-143(V)6/USQ

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the Interconnecting Group, ON-143(V)6/USQ, hereinafter referred to as the IG.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-C-5015 | Connectors, Electrical, Circular Threaded, AN Type, General Specification For |
| MIL-E-16400 | Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For |
| MIL-E-17555 | Electronic And Electrical Equipment, Accessories, And Provisioned Items (Repair Parts): Packaging Of |
| MIL-C-26482 | Connectors, Electrical, (Circular, Miniature, Quick Disconnect, Environment Resisting), Receptacles And Plugs, General Specification For |
| MIL-I-46058 | Insulating Compound, Electrical (For Coating Printed Circuit Assemblies) |
| MIL-C-55543 | Cable, Electrical Flat Multiconductor, Flexible, Unshielded |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-595 | Colors |

MILITARY

| | |
|---|---|
| MIL-STD-12 | Abbreviations For Use On Drawings, And In Specifications, Standards And Technical Documents |
| MIL-STD-105 | Sampling Procedures And Tables For Inspection By Attributes |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-003-12), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5820

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|---|---|
| MIL-STD-108 | Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment |
| MIL-STD-109 | Quality Assurance Terms And Definitions |
| MIL-STD-188-114 | Electrical Characteristics Of Digital Interface Circuits |
| MIL-STD-415 | Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For |
| MIL-STD-454 | Standard General Requirements For Electronic Equipment |
| MIL-STD-461 | Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference |
| MIL-STD-462 | Electromagnetic Interference Characteristics, Measurement Of |
| MIL-STD-471 | Maintainability Verification/Demonstration/Evaluation |
| MIL-STD-740 | Airborne And Structureborne Noise Measurements And Acceptance Criteria Of Shipboard Equipment |
| MIL-STD-781 | Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution |
| MIL-STD-810 | Environmental Test Methods And Engineering Guidelines |
| MIL-STD-965 | Parts Control Program |
| DoD-STD-1399, Section 070, Part 1 | Interface Standard For Shipboard Systems, D.C. Magnetic Field Environment (Metric) |
| DoD-STD-1399, Section 300 | Interface Standard For Shipboard Systems, Electrical Power, Alternating Current (Metric) |
| MIL-STD-1472 | Human Engineering Design Criteria For Military Systems, Equipment And Facilities |

HANDBOOKS

MILITARY

| | |
|---|---|
| MIL-HDBK-241 | Design Guide For Electromagnetic Interference (EMI) Reduction In Power Supplies |
| MIL-HDBK-251 | Reliability/Design Thermal Applications |

2.1.2 Other Government drawings and publications. The following other Government drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

SPACE AND NAVAL WARFARE SYSTEMS COMMAND (SPAWAR)

| | |
|---|---|
| 55910-0104137 | Circuit Card Assembly, Dynamically Adaptive Receiver Transmitter |
| 55910-0104140 | Circuit Card Assembly, Radio Interface Buoy |
| 55910-0104146 | Panel Assembly, Front |
| 55910-0104150 | Chassis Assembly |
| 55910-0104151 | Cage Assembly, Circuit Module |
| PL0104151 | Parts List, Cage Assembly Module |
| 55910-0104152 | Cage Assembly, BLACK Circuit Module |
| 55910-0104153 | Control Indicator, C-10582A/USQ |
| DL0104153 | Data List, Control Indicator, C-10582A/USQ |
| 55910-0104157 | Power Supply, 5 VDC Or ±12 VDC |
| 55910-0104160 | Interconnecting Group, ON-143(V)6/USQ |
| 55910-0104604 | Circuit Card Assembly, RED/BLACK Interface |
| 55910-0104642 | Panel, Power Distribution |
| 55910-0104645 | Panel Assembly, Rear |
| 55910-0105714 | Cable Assembly, BLACK I/O (1W1) |
| 55910-0105715 | Cable Assembly, BLACK I/O (1W2) |
| 55910-0105716 | Cable Assembly, BLACK I/O (1W3) |
| 55910-0105717 | Cable Assembly, NTDS Input (1W4) |
| 55910-0105718 | Cable Assembly, NTDS Output (1W5) |



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| | |
|---|---|
| 55910-0105719 | Cable Assembly, HSP (1W6) |
| 55910-0105720 | Cable Assembly, PTO (1W7) |
| 55910-0105721 | Cable Assembly, VOC, REP (1W8) |
| 55910-0105722 | Cable Assembly, KG, TTY (1W9) |
| 55910-0105723 | Cable Assembly, Jumper (1W10) |
| 55910-0105724 | Cable Assembly, Power Supply (1W11) |
| 55910-0105725 | Cable Assembly, SEN (1W12) |
| 55910-0105726 | Cable Assembly, Blower (1W13) |
| 55910-0105727 | Cable Assembly, RED/BLACK, PWR-SIG (1W14) |
| 89199-0008122 | Circuit Card Assembly, Processor Control 1 |
| 89199-0011097 | Circuit Card Assembly, Memory, Read Only, Reprogrammable 3 |
| 89199-0011101 | Circuit Card Assembly, Memory, Random Access (RAM 3) |

PUBLICATIONS

NATIONAL SECURITY AGENCY (NSA)

| | |
|---|---|
| NACSIM 5100A | Compromising Emanations Laboratory Test Standard Electromagnetic |
| NACSIM 5110 | Compromising Emanations Laboratory Test Standard Electromagnetic |

BUREAU OF NAVAL PERSONNEL (NAVPERS)

| | |
|---|---|
| NAVPERS 18068 | Manual Of Navy Enlisted Manpower And Personnel Classifications And Occupational Standards |

NAVAL AIR SYSTEMS COMMAND (NAVAIR)

| | |
|---|---|
| WS-6536D | Procedures And Requirements For Preparation And Soldering Of Electrical Connections |

SPACE AND NAVAL WARFARE SYSTEMS COMMAND (SPAWAR)

| | |
|---|---|
| EE130-AC-OMI-010/E 106 ON143V6 | Operation And Maintenance Instructions For Interconnecting Group (IG) ON-143(V)6 |
| NAVMAT P-4855.1 | Navy Power Supply Reliability |

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

| | |
|---|---|
| EIA RS-310-C-77 | Racks, Panels And Associated Equipment |

(Application for copies should be addressed to Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)



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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The IG shall be in accordance with MIL-E-16400, to the extent specified herein.

3.1.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Parts, materials, and processes. The design practices, parts, processes, and materials developed and procured for use in the IG shall ensure fulfillment of all requirements of this specification.

3.2.1 Selection. Selection of parts, materials, and processes shall conform to the Selection paragraph of MIL-E-16400.

3.2.1.1 Used or damaged parts or materials. Used or damaged parts or materials shall not be used.

3.2.1.2 Interchangeability. Interchangeability shall conform to the Interchangeability paragraph of MIL-E-16400. Drawing (DWG) 55910-0104160 shall be the standard for measuring interchangeability of all applicable units, assemblies, subassemblies, and parts.

3.2.2 Parts. Parts shall be as specified in 3.2.2.1 through 3.2.2.10.

3.2.2.1 Parts control. Parts control shall be in accordance with MIL-STD-965, Procedure I or II, as applicable.

3.2.2.2 Standard parts. Standard parts shall be in accordance with the Standard parts paragraph of MIL-E-16400.

3.2.2.3 Nonstandard parts. Nonstandard parts shall conform to the Nonstandard parts paragraph of MIL-E-16400.

3.2.2.4 Parts selection. The operating temperature of heat dissipating and heat sensitive parts shall be within the thermal stress identified in the reliability prediction report. Heat sensitive parts shall be thermally isolated from heat sources by one or more of the functions specified in a through d:

a. Physical separation
b. Use of thermal insulation techniques
c. Placement of such parts upstream in the cooling fluid or path, and heat sources downstream
d. Maximum use shall be made of conduction techniques to remove heat from heat dissipating parts along short metallic paths of large cross-sectional area to the heat sink or exchanger.

3.2.2.5 Derating of parts. Derating of electronic parts and materials shall conform to the Derating paragraph of MIL-E-16400.

3.2.2.6 Air filters. The forced air cooling inlet and outlet temperature differential shall not exceed 14° Celsius (C). Filters shall be provided and located at all air intakes. The filters shall be readily removable for cleaning without dissassembly of the equipment.



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3.2.2.7 Capacitors. Capacitors shall be selected in accordance with the Capacitors paragraph of MIL-E-16400.

3.2.2.7.1 Capacitors, fixed, dry electrolytic (aluminum foil). Capacitors, fixed, dry electrolytic (aluminum foil) shall be in accordance with the Capacitors, fixed, dry electrolytic (aluminum foil) paragraph of MIL-E-16400.

3.2.2.7.2 Capacitors, fixed, paper on paper-plastic (nonmetallic case). Fixed capacitors, paper or paper-plastic (nonmetallic case) shall be in accordance with the Capacitors, fixed, paper or paper-plastic (nonmetallic case) paragraph of MIL-E-16400.

3.2.2.7.3 Mica-dielectric capacitors. The use of mica-dielectric capacitors shall be in accordance with the Mica-dielectric capacitors paragraph of MIL-E-16400.

3.2.2.7.4 Capacitors, fixed, paper dielectric. Fixed paper dielectric capacitors shall not be used.

3.2.2.8 Microelectronic devices. Microelectronic devices shall conform to the Microelectronic devices paragraph of MIL-E-16400.

3.2.2.9 Connectors. Connectors shall conform to MIL-C-26482 or MIL-C-5015, as shown on DWG 55910-0104160.

3.2.2.9.1 Mating connectors. A complete set of mating halves of connectors, backshells, and protective caps shall be provided. Except for coaxial types, they shall be solderless (crimp-on) type whenever possible. Connectors shall be selected, designed, and installed so they cannot be mismatched or cross-connected.

3.2.2.10 Switches. All electrically actuated mechanical switches, except momentary switches, built-in test (BIT) and built-in test equipment (BITE) switches, and relays shall be of the self-latching type, and shall not require power to remain in any selected position.

3.2.3 Materials. All material used in the construction and fabrication of the IG shall be in accordance with the material requirements specified in MIL-E-16400. The requirement for nonmagnetic material is excluded. The contractor shall certify that the equipment materials used will not be of toxic or hazardous substances and that the material shall have no adverse effect on the health of personnel when used for its intended purposes.

3.2.3.1 Arc-resistant materials. Arc-resistant materials shall conform to the Arc-resistant materials paragraph of MIL-E-16400.

3.2.3.2 Insulating materials, electrical. Electrical insulating materials shall conform to the Insulating materials, electrical paragraph of MIL-E-16400.

3.2.3.3 Dissimilar metals. The selection and application of materials used in the design and fabrication shall conform to the Dissimilar metals paragraph requirements of MIL-E-16400.

3.2.3.4 Ferrous alloys, corrosion resistant. Ferrous alloys shall conform to the Ferrous alloys, corrosion resistant paragraph of MIL-E-16400.

3.2.3.5 Aluminum alloys. Aluminum alloys shall be in accordance with the Light weight metal (aluminum alloy) paragraph of MIL-E-16400.

3.2.3.6 Other metals. Non-ferrous metals, except aluminum, shall be in accordance with the Non-ferrous metals paragraph of MIL-E-16400.

3.2.3.7 Unacceptable materials. Equipment manufactured in accordance with this specification shall not include polychlorinated biphenyls, asbestos and asbestos compounds, fragile or brittle materials, beryllium and beryllium compounds unless so identified, and lithium and lithium compounds not specifically approved by the procuring activity. The Unacceptable materials paragraph of MIL-E-16400 shall apply.



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3.2.3.8 Adhesives. Adhesives shall conform to the Adhesives paragraph of MIL-E-16400.

3.2.3.9 Lubricants. Lubricants shall conform to the Lubricants paragraph of MIL-E-16400.

3.2.3.10 Fungus. The IG shall be resistant to fungi in both the operating and nonoperating condition and shall not be adversely affected when tested in accordance with the Fungus test paragraph of MIL-E-16400.

3.2.4 Processes. Processes shall be as specified in 3.2.4.1 through 3.2.4.4.2.

3.2.4.1 Soldering. Soldering shall conform to WS-6536D. Soft soldering shall not be used for mechanical support.

3.2.4.2 Structural welding. Structural welding shall conform to the Welding, structural paragraph of MIL-E-16400.

3.2.4.3 Castings. Castings shall be in accordance with the Castings paragraph of MIL-E-16400.

3.2.4.4 Painting. Only the exterior surface of the cabinet shall be painted.

3.2.4.4.1 Surface preparation. Surface preparation shall conform to the Painting paragraph of MIL-E-16400.

3.2.4.4.2 Color. Paint shall be gray enamel, Color No. 26307, in accordance with FED-STD-595. Preparation and painting shall conform to the Painting paragraph of MIL-E-16400.

3.3 Design. The IG shall be manufactured in accordance with DWG 55910-0104160 in order to produce IGs that are identical in form, fit, and function to existing IGs. The manufacturer shall provide IGs that are pin-for-pin compatible and interchangeable to the lowest replaceable unit (LRU) with existing IGs. Proposed changes, either to correct DWG discrepancies or to effect improvements, shall be approved by the procuring activity prior to incorporation.

3.3.1 IG description. The IG shall provide for satellite communications within one of the networks specified in a through c:

a. Officer-in-Tactical Command Information Exchange Subsystem (OTCIXS)
b. Submarine Satellite Information Exchange System (SSIXS)
c. Fleet Satellite Communications (FLTSATCOM) Secure Voice

The processor program shall provide the control required to interact with the operator and to allow the use of the same suite of equipment for the communications networks. Because the IG shall have a single radio channel capability, monitoring more than one network simultaneously will not be possible. The IG shall maintain TEMPEST isolation between RED/BLACK security classifications and perform the functions specified in d through j:

d. Generate timing delays
e. Time receiver synchronization
f. Provide vocoder synchronization
g. Provide message storage via the message queues specified in 1 through 6:
    1. OTCIXS transmit queue, 8K byte limit
    2. SSIXS transmit queue, 15K byte limit
    3. Report print queue
    4. Flash print queue
    5. Immediate print queue
    6. Retention queue
h. Initiate the self and system diagnostic tests specified in 1 through 6:
    1. Power-on BITE
    2. Crypto loop test
    3. SAT loop test



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4. Voice loop test
5. Data loop test
6. Control panel test

i. Provide the necessary low-level outputs for the teletypewriter (TTY) and the typing reperforator.

j. By means of a front-mounted control panel, allow the operator to select mode of operation, enter operating parameters, and verify and enter data by touching the appropriate indicator. For ease of operation, the front-mounted control panel shall be capable of being remotely located.

The IG shall be a standard 19-inch (in.) rack-mountable unit with an enclosed chassis. All controls and indicators necessary for operation shall be located on the front panel, although it shall be possible to remotely locate the control panel. A cover shall be provided for the IG when the control panel is in a remote location. All input, output, and power connectors shall be mounted on the rear panel. The IG front panel and chassis shall be mounted on slide rails to allow the IG to be extended from the case for access to the card cages and internal circuitry. The IG shall be broken down into the five basic areas specified in k through o:

k. Front panel, rear panel, and case
l. RED card cage
m. BLACK card cage
n. RED/BLACK interface
o. Power supplies

FIGURE 1 is an interface block diagram of the IG and system equipment. FIGURE 2 is an overall functional block diagram of the IG.

3.3.1.1 Front panel, rear panel, and case. The front panel shall contain the operating controls and indicators and the control panel. An internally-mounted fan and radio-frequency interference (RFI) attenuation air filter shall be used to cool the IG. When the front panel is mated with the housing, an RFI gasket on the front panel shall ensure RFI integrity. The front end chassis shall be capable of being extended by first loosening knurled fasteners at each corner and pulling outward on the handles on both sides of the front panel until the detents in the side rails engage. With the unit so extended, the RED and BLACK card cages, power supplies, and internal circuits shall be accessible. Flat cables with plug-in connectors shall provide the interface between the rear panel connectors and the IG circuitry.

3.3.1.2 RED card cage. The RED card cage shall be mounted to the chassis and shall be extended out of the case with the front panel. The RED card cage shall contain the 12 circuit cards that perform the various classified system functions. The RED card cage shall contain the cards specified in a through d:

a. One central processing unit (CPU)
b. Four random access memory (RAM)
c. Three read only memory (ROM)
d. Four dynamically adaptive receiver transmitter (DART):
    1. One vocoder DART
    2. One crypto DART
    3. One CPTTY DART
    4. One data link DART

Circuitry from the RED card cage to the rear panel shall be in the form of flat multiconductor ribbon cables. The RED card cage end of the ribbon cables shall terminate with 40-pin male strip plugs. The rear panel end of the ribbon cables shall terminate with round multipin connectors on the rear panel. The flexibility of the ribbon cables shall allow the chassis to be extended on the slide rails.



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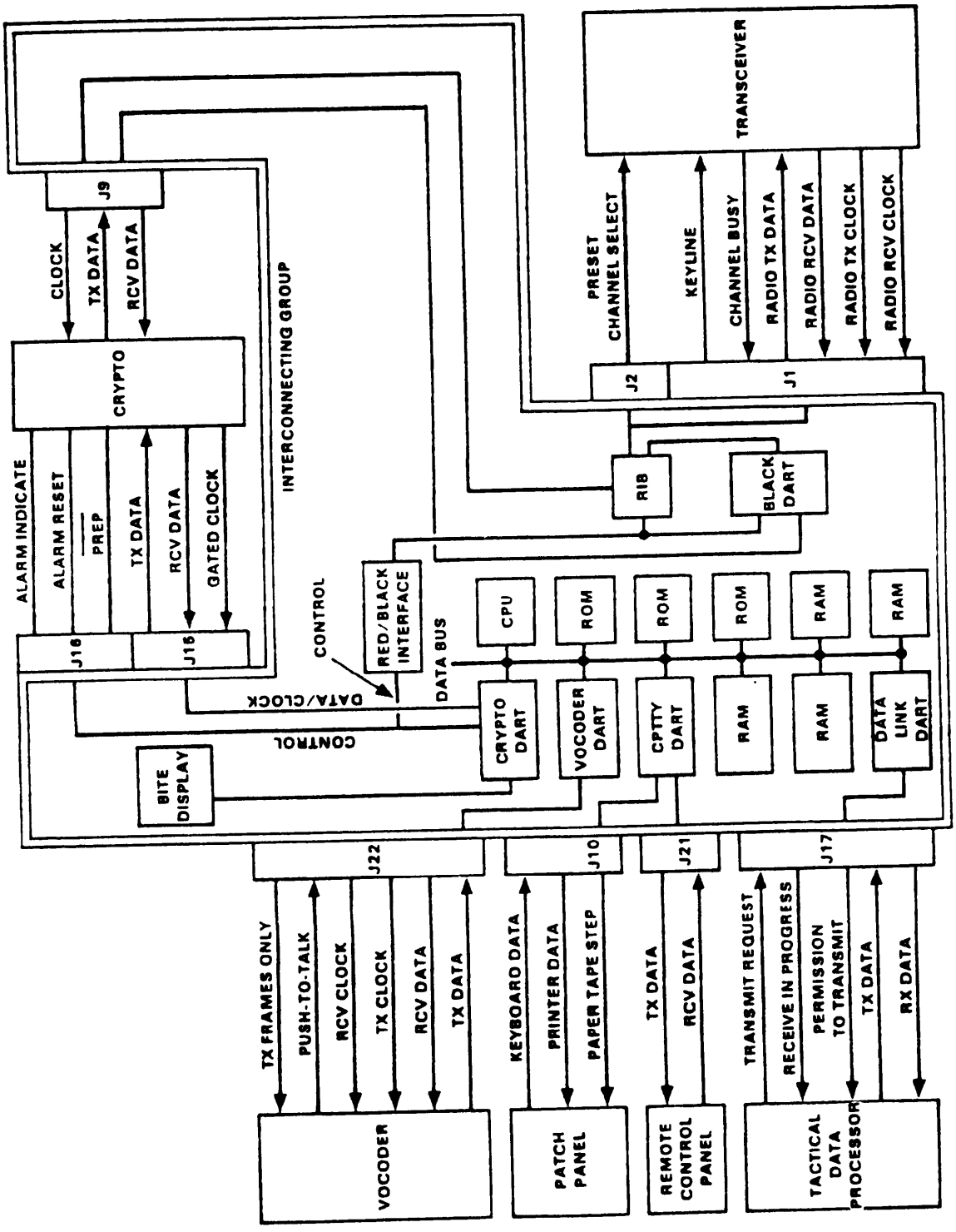



FIGURE 1. Detailed interface block diagram.



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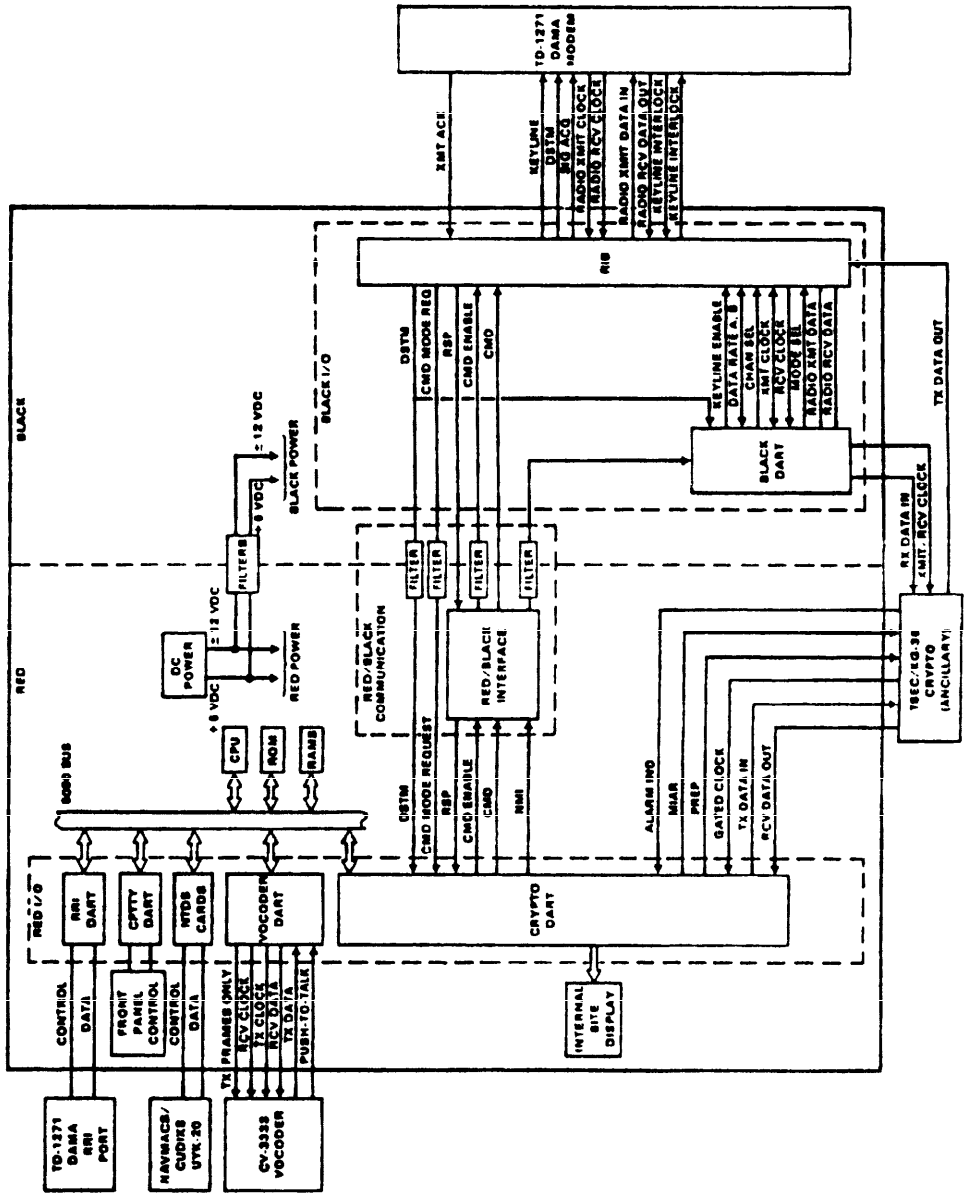



FIGURE 2. Interconnecting group, overall functional block diagram.



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3.3.1.3 BLACK card cage. The BLACK card cage shall be mounted to the front panel and slide in and out along with the RED card cage. The BLACK card cage shall contain two circuit cards that perform the various unclassified system functions. To provide TEMPEST isolation, the BLACK card cage shall be physically separated and electronically filtered from the RED card cage. Connections from the BLACK card cage to the rear panel shall be accomplished via ribbon cables as specified in 3.3.1.2. The BLACK card cage shall contain the cards specified in a and b:

a. One BLACK DART
b. One radio interface board (RIB)

3.3.1.4 RED/BLACK interface. Signals passing through the RED/BLACK interface card shall either be filtered by low-pass electrical filters or isolated by solid-state relays. The solid-state relays shall be located on the RED/BLACK interface card, housed on the RED chassis next to the RED/BLACK transition. The low-pass filters shall be chassis-mounted. All interconnecting wiring through the RED/BLACK partition shall be shielded.

3.3.1.5 Power supplies. The power supplies shall be located next to the RED card cage and shall supply the regulated +5 volts direct current (VDC) and ±12 VDC used for IG operation. Power supplies shall conform to or exceed the requirements specified in NAVMAT P-4855.1.

3.3.2 Interface definition. The IG shall interconnect the crypto, TTY, transceiver, audio encoder, and tactical data link processor in FLTSATCOM system installations as shown in FIGURE 1.

3.3.2.1 Data interface. The IG shall be capable of interfacing on the communications channels and at the data rates specified in TABLE I. All input/output (I/O) signals specified in TABLE I shall be in accordance with the Unbalanced voltage digital interface circuit paragraph of MIL-STD-188-114.

3.3.2.2 Electrical power interface. The IG shall operate from a single-phase, 60 hertz (Hz), 115-volt alternating current (VAC) power source which provides Type I power as specified in DoD-STD-1399, Section 300, and shall conform to the requirements and constraints of that document and NAVMAT P-4855.1. The IG shall not require more than 250 watts power.

TABLE I. IG I/O channels.

| (I/O) boards | Port ID 1/ | Description | Transfer data rate |
|---|---|---|---|
| Vocoder DART | J22 | Digital-coded voice | 2400 |
| CPTTY DART | J18 | TTY | 75 (selectable by strap option) |
| RIB | J1 | Data for radio transmission | 2400 |
| RIB | J2 | Remote Control, (AN/WSC-3) | N/A |
| BLK DART | J9 | Encrypted data to crypto | 2400 |
| RIB | J9 | Encrypted data from crypto | 2400 |
| Data link DART | J17 | Tactical data processor | 4800 |
| Crypto DART | J15/16 | KG-36 interface | 2400 |
| CPTTY DART | J21 | Remote control panel | 2400 |

1/ Identification



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3.3.2.2.1 Application of BLACK power. The IG shall be capable of direct application of unfiltered (BLACK) power without compromising emanations being present on the power line.

3.3.2.3 Strappable options. Provision shall be made for the selection of the options specified in a and b by backplane strapping (see 6.2):

a. Exclusion of SSIXS function. This option is appropriate for equipment to be installed onboard surface ships which will not require SSIXS capability.
b. Selection of TTY data rates 75 bits per second (bps), 300 bps, 1200 bps or 2400 bps

3.3.3 Major component list. The IG shall consist of the components specified in TABLE II. The component parts shall be fully interoperable and interchangeable with previcusly manufactured IG components to facilitate interoperability, interchangeability, training, and logistics.

TABLE II. Major component list.

| COMPONENT | DWG No. |
|---|---|
| Interconnecting Group (1) | 55910-0104160 |
| Panel Assembly, Front (1A1) | 55910-0104146 |
| Control Indicator, C-10582/USQ (1A1A1) | 55910-0104153 |
| Panel Assembly, Rear (1A2) | 55910-0104645 |
| Cage Assembly, BLACK Circuit Module (1A3) | 55910-0104152 |
| Cage Assembly, RED Circuit Module (1A4) | 55910-0104151 |
| Panel, Power Distribution (1A5) | 55910-0104642 |
| Circuit Card, RED/BLACK Interface (1A6) | 55910-0104604 |
| Circuit Card, BLACK DART (1A3A3) | 55910-0104137 |
| Circuit Card, RIB (1A3A6) | 55910-0104140 |
| Circuit Card, Vocoder DART (1A4A32) | 55910-0104137 |
| Circuit Card, CPTTY DART (1A4A30) | 55910-0104137 |
| Circuit Card, Data Link DART (1A4A29) | 55910-0104137 |
| Circuit Card, Crypto DART (1A4A28) | 55910-0104137 |
| Circuit Card, RAM (1A4A20) (C) | 89199-0011101 |
| Circuit Card, RAM (1A4A18) (A) | 89199-0011101 |
| Circuit Card, RAM (1A4A17) (9) | 89199-0011101 |
| Circuit Card, RAM (1A4A16) (8) | 89199-0011101 |
| Circuit Card, ROM (1A4A14) (2) | 89199-0011097 |
| Circuit Card, ROM (1A4A13) (1) | 89199-0011097 |
| Circuit Card, ROM (1A4A12) (0) | 89199-0011097 |
| Circuit Card, CPU (1A4A11) | 89199-0008122 |
| Power Supply (1PS1) | 55910-0104157-1 |
| Power Supply (1PS2) | 55910-0104157-2 |
| Cable Assembly (1W1) | 55910-0105714 |
| Cable Assembly (1W2) | 55910-0105715 |
| Cable Assembly (1W3) | 55910-0105716 |
| Cable Assembly (1W4) | 55910-0105717 |
| Cable Assembly (1W5) | 55910-0105718 |
| Cable Assembly (1W6) | 55910-0105719 |
| Cable Assembly (1W7) | 55910-0105720 |
| Cable Assembly (1W8) | 55910-0105721 |
| Cable Assembly (1W9) | 55910-0105722 |
| Cable Assembly (1W10) | 55910-0105723 |
| Cable Assembly (1W11) | 55910-0105724 |
| Cable Assembly (1W12) | 55910-0105725 |
| Cable Assembly (1W13) | 55910-0105726 |
| Cable Assembly (1W14) | 55910-0105727 |

3.3.4 Mechanical design. Mechanical design shall be as specified in 3.3.4.1 through 3.3.4.5.5.

3.3.4.1 Enclosures. The IG enclosures shall conform to the Enclosures paragraph of MIL-E-16400 and 3.3.4.2.

3.3.4.2 Degree of enclosure. The IG enclosure shall be dripproof (45 degrees) in accordance with MIL-STD-108.



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3.3.4.3 Equipment slide and tilt mechanisms. The equipment shall have slide mechanisms which shall be of suitably rugged construction to support the drawer in the withdrawn position. The slide mechanism shall slide with a minimum of sliding friction without a tendency to bind. The slide mechanism shall be constructed of corrosion-resistant material, such as corrosion-resistant steel. Guide support shall be provided between the rear of each chassis and the enclosure to prevent possible damage, due to shock, to the chassis slides.

3.3.4.4 Shock mounting. All slide-mounted equipment shall include shock pins at the rear of the unit.

3.3.4.5 Controls, indicators, and panel layouts. The design of controls and indicators, and their arrangement on operator and maintainer panels, shall conform to the criteria of MIL-STD-1472 and shall be in accordance with the Controls, indicators, and panel layout paragraph of MIL-E-16400. The color coding and marking of pin jacks and other test points shall be in accordance with MIL-STD-415. Abbreviations used on panels shall be in accordance with MIL-STD-12.

3.3.4.5.1 Front panel layout. Front panel layout of cabinets and drawers shall take into consideration sequencing procedures to facilitate performance by Navy electronics technicians, second class (ET-2) for set up, operation, and maintenance. All controls and metering necessary for normal operation of the IG shall be provided on the IG front panel.

3.3.4.5.2 Alarm lamps. Lamps used on front panels for alarms and other purposes shall conform to the requirements of the Transilluminated Displays paragraph of MIL-STD-1472. A capability for remote indication of the status of the alarm lamps shall be provided. Legend lights shall be used in preference to simple indicator lights except where design considerations demand that simple indicators be used. Lamps used in legend switches shall conform to the requirements of the Legend Switches paragraph of MIL-STD-1472. Legend switches shall be used in preference to toggle switches except where design considerations demand that toggle switches be used. Lamp testing shall be in accordance with the Lamp Testing paragraph of MIL-STD-1472.

3.3.4.5.3 Protection from improper operation. The IG design shall include features to protect the equipment from improper operational procedures and practices.

3.3.4.5.4 Chain reaction failures. To the extent practicable, the IG protective design shall inhibit chain reaction failures in which a failure in one component results in damage to other components.

3.3.4.5.5 Locking devices. Locking devices shall be provided on all critical internal adjustment controls in order to prevent the need for readjustment of controls after sustained periods of shock and vibration.

3.3.5 Airborne and structureborne noise. The IG shall conform to the requirements of MIL-STD-740 for Grade A, Type 3 equipment and as specified herein. The maximum allowable sound pressure level for airborne noise shall be 60 decibels amperes measured at 1 meter distance from the equipment. The structureborne noise criteria shall be 65 decibels (dB) for one third wave band frequencies up to 800 Hz. Both the airborne and structureborne noise criteria shall be applicable when the equipment is operating normally.

3.3.6 Thermal design. The thermal design for the IG shall be as specified in 3.3.6.1 and 3.3.6.2 and in accordance with the Thermal design paragraph of MIL-E-16400. Air cooling shall be used. The design of the equipment shall allow installation in open racks.

3.3.6.1 Heat removal. The heat removal method shall remove at least 80 percent of the heat generated within the equipment. MIL-HDBK-251 may be used as guidance in the design and selection of the heat removal method.

3.3.6.2 Cooling during maintenance. Cooling during maintenance shall conform to the Cooling during maintenance paragraph of MIL-E-16400.



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3.3.7 Prime power design. Prime power design shall be as specified in 3.3.7.1 through 3.3.7.6.2.

3.3.7.1 Performance. The IG shall conform to the requirements of 3.5 when operated as specified in 3.3.7.2 through 3.3.7.6.2, in accordance with DoD-STD-1399, Section 300, Type I. Power supply reliability shall conform to the requirements of NAVMAT P-4855-1.

3.3.7.2 Prime power voltage (low level equipment). Prime power voltage for low level equipment shall be 115 VAC, 60 Hz, single-phase.

3.3.7.3 Prime power control. A prime power circuit breaker shall be provided in the IG cabinet. The circuit breaker shall conform to the Circuit breakers paragraph of MIL-E-16400.

3.3.7.4 Corona and electrical breakdown prevention. Corona and electrical breakdown prevention shall conform to the Corona and electrical breakdown prevention paragraph of MIL-E-16400.

3.3.7.4.1 Insulation resistance. Insulation resistance shall conform to the Insulation resistance paragraph of MIL-E-16400.

3.3.7.5 Bonding and grounding. Bonding and grounding shall conform to the Bonding and grounding paragraph of MIL-E-16400.

3.3.7.6 Internal wiring and cabling. Internal wiring and cabling shall be in accordance with the Internal wiring and cabling paragraph of MIL-E-16400.

3.3.7.6.1 Flat multiconductor cable. Flat multiconductor cable shall be selected from MIL-C-55543.

3.3.7.6.2 Internal wiring hookup. The selection and application of internal electrical hookup shall conform to MIL-STD-454, Requirement 20.

3.3.8 Printed wiring. Printed wiring shall conform to the Printed wiring paragraph of MIL-E-16400.

3.3.8.1 Printed wiring board dimensions. Printed wiring board dimensions shall be specified in DWG 55910-1014160.

3.3.8.2 Conformal coating. Printed circuit (PC) cards shall be conformal coated to conform to MIL-I-46058.

3.3.9 Maintenance design. Maintenance design shall conform to the Maintenance design paragraph of MIL-E-16400.

3.3.9.1 Self-test provisions. The IG shall use BIT and BITE, or both, to:

a. Verify that the on-line equipment is operating properly
b. Fault located to the subassembly level when the equipment is on-line or off-line

Maintenance assistance modules may supplement BIT and BITE.

3.3.9.2 Accessibility. The IG shall be designed to provide accessibility to all limited life components. The IG shall conform to the Accessibility paragraph of MIL-E-16400.

3.3.9.3 Maintenance concept. The maintenance concept for correction of equipment malfunctions at the organizational level shall be the removal and replacement of LRUs to restore the equipment to specification requirements. LRUs removed at this maintenance level will be limited to plug-in modules, PC cards, or chassis-mounted parts which are readily isolated by BITE and standard shipboard general purpose test equipment. Complexity of LRUs should be kept to a level such that LRU repair can be accomplished at the intermediate maintenance level by Navy ET-2 personnel with skills no greater than those specified in NAVPERS 18068 for an ET-2.



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3.3.9.4 Modular construction. Modular construction shall conform to the Modular construction paragraph of MIL-E-16400.

3.3.9.5 Circuit test point design criteria. Circuit test point accesses shall be provided which conform to the criteria specified in MIL-STD-415.

3.3.9.6 Special tools. The IG shall be serviceable with a minimum of special tools (see 6.5). All special tools needed shall be supplied. Special tools shall be subject to approval by the Government and shall conform to the Special tools paragraph of MIL-E-16400.

3.3.9.7 Power receptacles. Convenience power receptacles shall not be provided.

3.3.9.8 Hinged covers, panels, and access doors. Hinged covers, panels, and access doors shall conform to the Hinged covers, panels, and access doors paragraph of MIL-E-16400.

3.3.9.9 Part protection. The IG shall be so designed that parts shall conform to the Part protection paragraph of MIL-E-16400.

3.3.9.10 Protective caps. Externally mounted connector receptacles which are not in continuous use shall be provided with protective caps in accordance with the Protective caps paragraph of MIL-E-16400.

3.4 Transportability. After preparation for delivery as required by Section 5, the IG or modules, or both, shall be transportable by air, rail, ship, or surface carrier, without damage.

3.5 Performance characteristics. Performance characteristics shall be as specified in 3.5.1 through 3.5.2.2.

3.5.1 Functional description. The IG is an integral part of any OTCIXS or SSIXS configuration (ashore, shipboard, or submarine). The IG performance, in its selected configuration, is controlled by both processor programs and IG electronic circuitry. The functional description specified in 3.5.1.1 through 3.5.1.8 cover the major functions and signal flow within the IG. A more detailed description to the subassembly and circuit card level is specified in the Operation and Maintenance Instructions EE130-AC-OMI-010/E 106 ON143V6.

3.5.1.1 General. The IG shall provide the means for performance of the seven major system functions specified in a through g:

a. Link control
b. Operator interface
c. Message data handling
d. Signal interfacing
e. Automated event sequencing
f. Secure voice operation
g. Loopback and BITE testing.

3.5.1.2 Link control. The link control function shall exercise the link protocol pattern established to conform to the operational needs of the subscribers based on the mode of operation selected by the operator. The IG shall automatically select one of the four satellite channels specified in a through d that directly correspond to the operator selected mode of operation:

a. OTCIXS
b. SSIXS
c. FLTSATCOM 1
d. FLTSATCOM 2

The OTCIXS and SSIXS channels utilize a link protocol that establishes the accessibility of the link with regard to the system mode, the relative priorities of the individual subscribers, and the precedence of message traffic. FLTSATCOM 1 and FLTSATCOM 2 channels are unstructured. OTCIXS and SSIXS each have their own link control (network cycle).



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3.5.1.3 Operator interface. The operator interface function shall be exercised via the operator control panel in the IG. By selecting distinct modes (Pages) on the control panel, the operator shall be able to enter commands and parameters for controlling system operation. These modes shall be as specified in the Operation and Maintenance instructions EE130-AC-OMI-010/E 106 ON143V6, Chapter 2, for the organizational level. The control panel shall display systems status information and alerts, and provide error or fault indications for both system level and IG level troubleshooting.

3.5.1.4 Message data handling. Message data handling encompasses the three subfunctions specified in a through c:

a. Message screening
b. Queueing
c. Storage

The subfunctions shall be accomplished by operator communication with the CPU and its program, in conjunction with automatic IG control of system equipment.

3.5.1.5 Signal interfacing. Components interfacing with the IG generate and require signals with differing characteristics. The IG shall convert these signals to the required form for system compatibility.

3.5.1.6 Automated event sequencing. The automated event sequencing function shall provide for automatic system operation by means of program controlled gating and timing logic.

3.5.1.7 Secure voice. The IG shall provide the means for the system secure voice capability. This shall be accomplished by interfacing the Audio Converter (Vocoder), CV-3333 digitized voice data with the Crypto, TSEC/KG-36, and the Transceiver, AN/WSC-3.

3.5.1.8 Loopback and BITE testing. The IG shall provide CPU controlled testing to isolate system or component faults. The capability to perform system transmit and receive tests through the use of a loopback mode shall be included. In this mode a test signal shall be transmitted and received over a designated path. The loopback mode shall include crypto loop test, voice loop test, and data loop test. The IG shall have BITE to assist in the detection and isolation of IG faults and interface failures.

3.5.2 Signal flow and internal functions. The IG shall be divided into two general areas. One area shall provide processing of RED (classified nonenciphered) data and the other shall provide processing of BLACK (enciphered) data. Units that will interface with the RED data area of the IG are the vocoder; Keyboard Send-Receive Set (TTY), AN/UGC-77; Typing Reperforator, TT-253/UG; Transmitter Distributor (TD), TT-187B/UG; and Tactical Data Processor (TDP). All TTY related equipment must be low level. The RED I/O port of the Crypto, TSEC/KG-36 shall also interface with the IG RED data area. Units that shall interface with the BLACK data area of the IG are the Transceiver, AN/WSC-3, the Buoy Interface Unit, and the BLACK I/O port of the crypto. All signals in or out of the IG shall be controlled at the control panel.

3.5.2.1 Basic transmit flow. For system transmission the IG shall receive signals input from the TTY, TD, TDP, or vocoder. A transmit (XMT) input signal shall initially enter a directly associated DART card (data link, vocoder, or CPTTY DART) for processing and interfacing with a common data bus. The bus shall connect these signals with an internal data processor consisting of a CPU, ROM, and RAM. Through programming, timing, and control, the processor shall transfer the signals to the crypto RED data input via the crypto DART. After the data is enciphered by the crypto, the data shall be fed back to the IG appearing as BLACK data at the BLACK DART. The RIB control circuit card shall then feed the BLACK data to the transceiver for transmission.

3.5.2.2 Basic receive flow. When the transceiver receives signals (RCV) designated for the vocoder, TTY printer, TDP, or reperforator, the transceiver forwards these signals (RCV) to the RIB card. The signals shall then follow a reverse flow from that of the XMT signals. The RIB shall forward the signals to the BLACK data input of the crypto through the BLACK DART. The signals shall be deciphered by the crypto and fed out as RED data to the crypto DART, then through the data bus to the specific DART for the interface with either the reperforator, TTY printer, TDP, or vocoder.



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3.6 Human engineering. The contractor shall conform to the human factors operational and maintenance design criteria of MIL-STD-1472 to achieve effective man-machine integration for both operation and maintenance.

3.7 Safety. Safety shall be in accordance with the Safety (personnel hazard) paragraph of MIL-E-16400.

3.7.1 Leakage current. The IG leakage current shall not exceed 5 milliamperes.

3.8 Physical characteristics. Physical characteristics shall be in accordance with 3.8.1 and 3.8.2.

3.8.1 Dimensions. The dimensions of the IG shall permit mounting in a standard 19-in. rack as specified in EIA RS-310-C-77. Dimensions shall conform to DWG 55910-0104150.

3.8.2 Weight. The weight of the IG, including the control panel, shall not exceed 73 pounds. The IG shall be prominently labeled in accordance with the weight labeling requirements of MIL-STD-1472.

3.9 Environmental service conditions. The IG shall be capable of continuous reliable operation under the environmental conditions specified in 3.9.1 through 3.9.9. The IG shall operate within tolerances throughout the tests specified herein, without alignment or adjustment, other than accessible controls on equipment front panels.

3.9.1 Temperature. Temperature shall be as specified in 3.9.1.1 and 3.9.1.2.

3.9.1.1 Operating temperature. The IG shall conform to the requirements specified herein when operating over a temperature range of -0° C to +50° C.

3.9.1.2 Nonoperating temperature. The IG shall not be damaged, nor shall specified performance be degraded, when restored to the operating temperature range after being subjected to the nonoperating temperature range of -62° C to +71° C.

3.9.2 Altitude (nonoperating). The IG shall conform to requirements of MIL-STD-810, Method 500.2, Procedure I.

3.9.3 Shock. The IG shall conform to the Shock paragraph of MIL-E-16400.

3.9.4 Vibration. The IG shall conform to the Vibration paragraph of MIL-E-16400. The upper vibration frequency limit shall be 50 Hz.

3.9.5 Magnetic field. The IG shall conform to the magnetic field environmental requirements of DoD-STD-1399, Section 070, Part 1, Interface characteristics paragraph.

3.9.6 Inclination. The IG shall conform to the Inclination test limits (submarine installations) paragraph of MIL-E-16400 to a maximum angle of 45 degrees.

3.9.7 Dust (fine sand). The IG shall conform to the requirements of MIL-STD-810, Method 510.2 and MIL-E-16400.

3.9.8 Humidity. The IG shall conform to the Humidity paragraph of MIL-E-16400.

3.9.9 Salt fog. The IG requirements for salt fog shall be in accordance with the Sheltered equipment paragraph of MIL-E-16400.

3.10 Electromagnetic environmental effects ($E^3$). The IG shall conform to the $E^3$ requirements specified in 3.10.1 through 3.10.4.

3.10.1 Electromagnetic interference (EMI). The IG shall conform to the requirements of MIL-STD 461, for Class A4 equipments when tested in accordance with the test methods of MIL-STD-462 for digital equipment.



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3.10.2 Electromagnetic compatibility (EMC) guidance. MIL-HDBK-241 may be used for EMI reduction in power supplies.

3.10.3 Compromising emanations (TEMPEST). The IG shall be capable of withstanding test of NACSIM 5100A and NACSIM 5110 to include electrical field test, magnetic field test, power line conduction test, and signal or control line conduction test.

3.10.4 Electromagnetic pulse (EMP). All equipment interface pins and terminals of leads, including grounds and neutrals, shall not exhibit any permanent malfunction, degradation of performance, or deviation from specified indications after being subjected to a test signal having either the waveform and common mode current level shown in FIGURE 3 or the waveform and common mode voltage level shown in FIGURE 4, whichever occurs first at the specific test frequency.

3.11 ID and marking. The IG shall be identified and marked in accordance with MIL-STD-454, Requirement 67.

3.12 Government-furnished property (GFP). The GFP (see 6.4) specified in 3.12.1 and 3.12.2 is available to the contractor.

3.12.1 Software. The Government shall provide a ROM board for programming the DART or ROM boards which includes a program for the BITE.

3.12.2 Equipment. The Government shall provide the equipment specified in a through f:

a. Two crypto simulators
b. One TTY (AN/UGC-20, or equivalent)
c. One set of master patterns for PC boards
d. One ultra-high frequency tactical satellite communications buoy
e. DWG 55910-101-4160
f. One IG.

3.13 Quantitative reliability requirements. The upper test mean-time-between-failures (MTBF) ($O_0$ as defined by MIL-STD-781) shall be 4000 hours for the IG. The lower test MTBF ($O_1$ as defined by MIL-STD-781) shall be 2000 hours. The MTBF times shall apply for Category 3A, sheltered shipboard equipment, under the conditions specified herein.

3.14 Maintainability. Maintainability shall be as specified in 3.14.1 through 3.14.3.

3.14.1 Quantitative organizational corrective maintenance requirement. The mean-time-to-repair shall not exceed 30 minutes and the maximum-corrective-maintenance-time ($M_{maxct}$) at the 90th percentile shall not exceed 90 minutes when corrective maintenance is accomplished at the organizational level by the replacement of lowest subassemblies (modules, printed circuit boards) and chassis-mounted components (including electronic, electrical, electromechanical and mechanical parts). The corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all maintenance tasks. The IG is considered in a corrective maintenance mode whenever a failed condition or stoppage of any individual operation occurs.

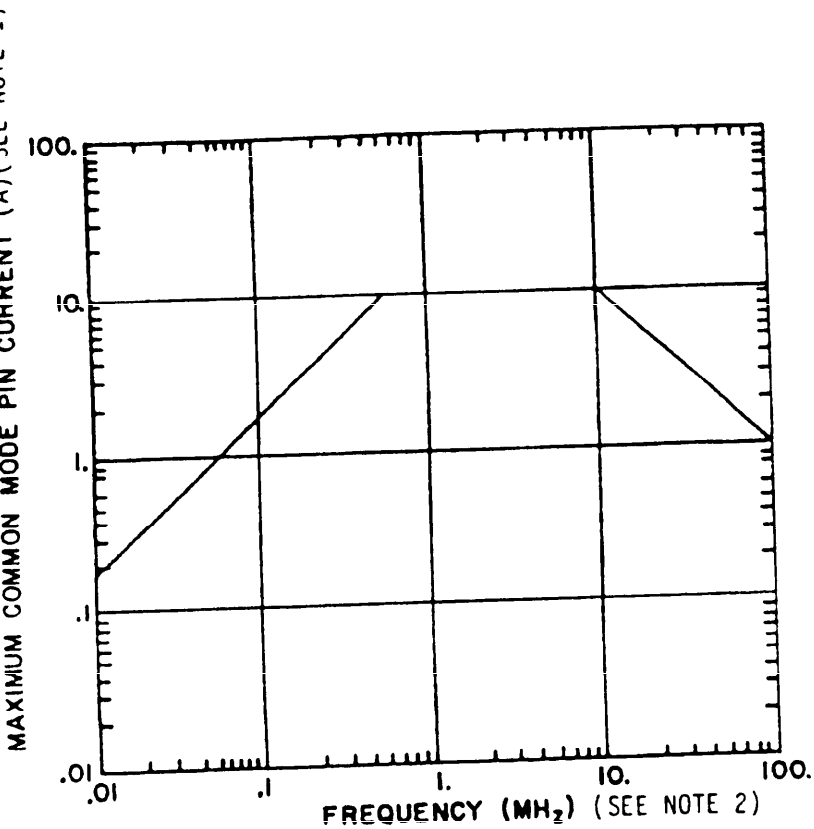
3.14.2 Preventive maintenance during missions. The only preventive maintenance shall consist of cleaning the IG air filter, cleaning of control panel touch switches, and inspecting the IG interior. The design shall facilitate the removal and installation of the air filter so that the filter can be removed, cleaned, and installed within an elapsed time of no more than 12 minutes. Access to the RED and BLACK card cages shall be such that the cabinet case can be opened, circuit cards and wiring harnesses inspected, and the cabinet closed within a period of 6 minutes.

3.14.3 Preventive maintenance between missions. The IG shall be designed to have a maximum preventive-maintenance downtime of 4 hours between missions, with no single task requiring more than 2 hours.

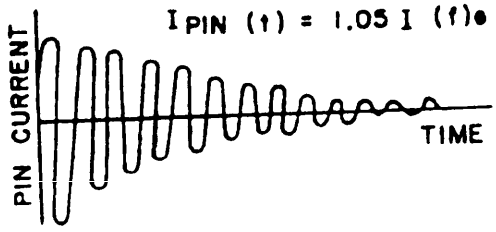
3.15 Workmanship. Workmanship shall be as specified in 3.15.1 and 3.15.2.



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$$I_{PIN}(t) = 1.05\, I(f)\, e^{-0.2ft} \sin(2\pi ft)$$

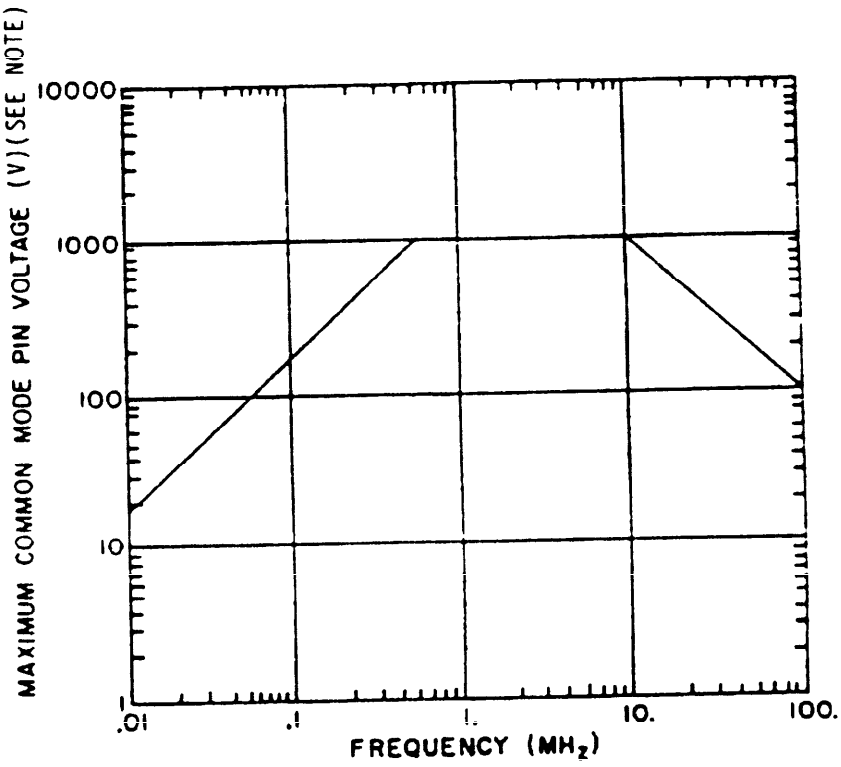


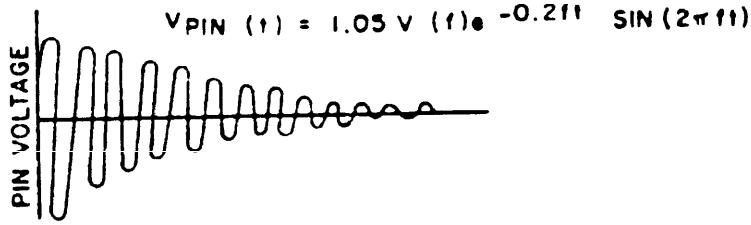

NOTES:
1. Amperes
2. Megahertz

FIGURE 3. Interface PIN current.



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NOTE: Volts

FIGURE 4. Interface PIN voltage.



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3.15.1 General workmanship. Workmanship shall conform to the Workmanship paragraph of MIL-E-16400.

3.15.2 Workmanship screen. All IGs shall withstand a defect detection vibration screen of random type vibration (see 4.8.2.1) at $0.04g^2/Hz$ ±3 dB from 80 Hz to 350 Hz and temperature cycling (see 4.8.2.2) with operation at the maximum and minimum operating temperatures and dwell time at the nonoperating temperature extremes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
   1. Production inspection (Group A) (see 4.4.1)
   2. Production control inspection (Group B) (see 4.4.2)
   3. Environmental inspection (Group C) (see 4.4.3)
c. Inspection of preparation for delivery (see 4.9)

4.3 First article inspection. The number of units to be submitted for first article inspection shall be as specified in the contract (see 6.2). First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE III.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.4.

4.4.1 Production inspection (Group A). Production inspection shall be conducted on each IG offered for delivery. Production inspection shall comprise such examinations and tests which will prove the workmanship and reveal the omissions and errors of the production process, such as functional and performance tests at a limited number of points; tests which detect deviation from design; tests of adjustment; and tests which detect hidden defects of material. Production inspection shall include the examinations and tests specified in Group A of TABLE III.



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TABLE III. Examinations and tests.

| Examination or test | Requirement paragraph | Test paragraph | First article inspection | Quality conformance inspection Group A | Group B | Group C |
|---|---|---|---|---|---|---|
| Interface | 3.3.2 | 4.6.3, 4.6.4 | X | X | | |
| Steady-state voltage and frequency | 3.3.2.2 | 4.6.1 | X | | X | |
| Transient voltage | 3.3.2.2 | 4.6.1 | X | | | |
| Transient frequency | 3.3.2.2 | 4.6.1 | X | | | |
| Spike voltage | 3.3.2.2 | 4.6.1 | X | | | |
| Power interruption | 3.3.2.2 | 4.6.6 | X | | X | |
| Power and power factor | 3.3.2.2 | 4.6.6 | X | | X | |
| Harmonic current | 3.3.2.2 | 4.6.1 | X | | | |
| Enclosures | 3.3.4.1 | 4.6.1 | X | | X | |
| Thermal design | 3.3.6 | 4.6.5 | X | | | |
| Maintainability demonstration | 3.3.9, 3.14 | 4.7 | X | | | |
| Preperformance | 3.5 | 4.6.1 | X | | X | X |
| Safety | 3.7 | 4.6.1 | X | X | | |
| Temperature (low and high) | 3.9.1 | 4.6.1 | X | | | X |
| Altitude | 3.9.2 | 4.6.1 | X | | | |
| Shock | 3.9.3 | 4.6.8 | X | | | X |
| Vibration | 3.9.4 | 4.6.9 | X | | | X |
| Magnetic field | 3.9.5 | 4.6.1 | X | | | |
| Inclination | 3.9.6 | 4.6.1 | X | | | X |
| Dust (fine sand) | 3.9.7 | 4.6.1 | X | | | |
| Airborne and structureborne noise | 3.3.5 | 4.6.1 | X | | | X |
| Leakage current | 3.7.1 | 4.6.7 | X | | | |
| Humidity | 3.9.8 | 4.6.1 | X | | | |
| Salt fog | 3.9.9 | 4.6.1 | X | | | |
| Fungus | 3.2.3.10 | 4.6.1 | X | | | |
| EMI | 3.10.1 | 4.6.10 | X | | | X |
| TEMPEST | 3.10.3 | 4.6.11 | X | | | |
| Surface examination | | | | | | |
| Parts, materials, and processes | 3.2 | 4.6.1 | X | X | | |
| Dimensions | 3.8.1 | 4.6.1 | X | X | | |
| Weight | 3.8.2 | 4.6.1 | X | X | | |
| Weight labeling | 3.8.2 | 4.6.1 | X | X | | |
| Finish | 3.2.4.4 | 4.6.1 | X | X | | |
| Identification and marking | 3.11 | 4.6.1 | X | X | | |
| General workmanship | 3.15.1 | 4.8.1 | X | X | | |
| EMP | 3.10.4 | 4.6.10.1 | X | | | |
| Workmanship screen | 3.15.2 | 4.8.2 | | X | | |
| Interchangeability | 3.2.1.2 | 4.6.2.1, 4.6.2.2 | X | | | |



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4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis in accordance with the inspection procedures of MIL-STD-105 to an acceptable quality level of 6.5 percent using the special inspection level S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on IGs that have passed production inspection. Production control inspection shall consist of the examinations and functional and performance tests throughout the entire range of operation; detection of any deterioration of the design by wear of dies, molds, and jigs, by variations in parts; the recording of deviations in the processing temperature rise during equipment operation and ability of the equipment to withstand this heat; tests of efficiency; and tests of the system performance. These tests shall be performed on the complete IG as offered for delivery. Production control inspection shall include the examinations and tests specified in Group B of TABLE III.

4.4.3 Environmental inspection (Group C). Environmental inspection shall encompass tests of equipment durability, simulated service tests; tests of the effects of environment (such as extremes of temperature and humidity, effect of salt fog); effects of shock, vibration, and inclination. Environmental inspection shall include the examinations and tests specified in Group C of TABLE III. Environmental inspection shall be performed on a sampling basis (see 4.4.3.1) on units which have been subjected to, and have passed, production inspection.

4.4.3.1 Environmental inspection samples. Samples for environmental inspection shall be selected without regard to quality in accordance with a and b:

a. One sample from the first 50 units fabricated
b. One sample from each successive 100 units, or fraction thereof, produced

4.4.4 Noncompliance. If a sample fails to pass the inspection specified in 4.4.3, the manufacturer shall notify the contracting officer and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the contracting officer has been taken. After the corrective action has been taken, Group C inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the contracting officer). Group A and Group B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the Group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the contracting officer.

4.5 Test equipment and test conditions. Test equipment and test conditions for first article inspection and quality conformance inspection shall conform to the quality assurance provisions of MIL-E-16400 specified in a through c:

a. Test facilities, chambers, and apparatus
b. Standard test conditions
c. Performance of test

4.6 Test methods. Test methods shall be as specified in 4.6.1 through 4.6.11.

4.6.1 Examinations and tests. Examinations and tests specified in TABLE III shall be conducted in accordance with the applicable test methods of MIL-E-16400, except as specified in 4.6.2 through 4.6.11.

4.6.2 Interchangeability. Interchangeability shall be as specified in 4.6.2.1 and 4.6.2.2.

4.6.2.1 RAM and DART board interchangeability test. The equipment shall be tested with each of the four DART boards in each slot provided for DART boards. That is, install DART board 1 in slot 1, DART board 2 in slot 2, DART board 3 in slot 3, and DART board 4 in slot 4. Test the equipment for performance. Repeat the test with DART board 1 in slot 2, DART board 2 in slot 3, DART board 3 in slot 4, and DART board 4 in slot 1. Continue testing until each board has been tested in each slot. Perform the same test for the three RAM boards in the three slots provided.



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4.6.2.2 LRU interchangeability. Each IG LRU manufactured under this specification shall be tested for interchangeability with the Government-furnished LRUs. The interchangeability test shall be conducted by interchanging the new LRUs with those of the Government-furnished IG in every combination, and testing each IG for proper operation. The test shall be conducted in accordance with a contractor-furnished and Government-approved interoperability test plan.

4.6.3 Interface test. The BITE testing shall be conducted with the interfacing equipment shown in FIGURE 1 to demonstrate compliance with 3.3.2.

4.6.4 SSIXS data link. The SSIXS data link shall be tested by the BITE.

4.6.5 Thermal design. The Thermal performance test of MIL-E-16400 shall be performed while the IG is undergoing the first article temperature test.

4.6.5.1 Thermal survey. A thermal survey shall be performed on those parts dissipating 1 percent, or more, of the total power dissipation of the unit in which the part is located and to ensure that heat-sensitive parts are within the thermal stress identified in the reliability prediction report and the part derating requirement. The thermal data shall include, but not be limited to, the actual operating temperature of the part and any visible evidence of deterioration of the equipment.

4.6.5.2 Survey conditions. The survey shall be conducted in accordance with the Test condition paragraph of MIL-E-16400.

4.6.6 Electrical power tests. Tests shall be conducted in accordance with MIL-E-16400 and DoD-STD-1399, Section 300 to ensure compliance with requirements of 3.3.2.2 and 3.3.7. Loss of memory contents is acceptable during power interruption testing.

4.6.7 Leakage current test. The leakage current test shall be as specified in 4.6.7.1 and 4.6.7.2.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE IG DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES WITHOUT ADEQUATE ELECTRIC SHOCK PROTECTION.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.6.7.1 IG test connections. After power removal each IG shall be placed on an insulated surface. The safety ground conductor between the IG and the source power shall be opened during the test. The IG shall be connected as shown in FIGURE 5 if it is connected to single-phase source power.

4.6.7.2 Measurement. Leakage current shall be measured on the IG in its normal operating configuration. IG controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true root-mean-square (rms) voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 V at the highest nominal power line voltage and the higher and lowest nominal power line frequencies for which the IG is designed. The overall measurement error shall not exceed 5 percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed. The voltage shall be measured from each part of ground and from each part to all other simultaneously accessible parts for every combination of switch positions available in FIGURE 5. The open safety ground conductor shall be reconnected immediately after the test is completed.



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4.6.8 Shock. The IG shall be shock tested in accordance with the Shock paragraph of MIL-E-16400 to determine conformance to the requirements of 3.9.3. The unit under test shall be operated before, during, and after the shock test. It is acceptable that circuit breakers may be tripped or that some loss of memory contents results from the shock impact, however, the IG shall be capable of normal operation after re-initialization procedures following the shock test.

4.6.9 Vibration. The IG shall be subjected to the test specified in the Environmental vibration paragraph of MIL-E-16400 to determine conformance to 3.9.4.

4.6.10 EMI tests. EMI, emission, and susceptibility requirements shall be verified by tests performed in accordance with MIL-STD-462.

4.6.10.1 EMP test. The requirements of 3.10.4 shall be verified by the test methods specified in a through d:

a. Applicable test frequencies. The tests shall be performed for the injection frequencies of 0.01 MHz, 0.1 MHz, 0.5 MHz, 1 MHz, 2 MHz, 5 MHz, 10 MHz, 20 MHz, 50 MHz, and 100 MHz. In addition, the test shall be performed at the test sample's critical frequencies, such as local oscillator, power switching frequencies, and harmonics, radio frequencies, clock frequencies, bandpass frequencies, and so forth as specified in the EMI test plan.

b. Apparatus. Test apparatus shall consist of the items specified in 1 through 4:

1. Source injection pulse generator with the capabilities specified in A through D:

A. Damped sinusoidal transient as shown in the applicable limit and as described by the equations specified in (1) and (2):

(1) $I_{pin(t)} = 1.05\ I(f)e^{-0.2\ ft}\ SIN\ (2\pi\ ft)$

(2) $V_{pin(t)} = 1.05\ V(f)e^{-0.2\ ft}\ SIN\ (2\pi\ ft)$

where:

$I_{pin(t)}$ = Pin to case current, in A at time, t

$V_{pin(t)}$ = Pin to case voltage, in V at time, t

f = Specific test frequency, in MHz

t = Time, in seconds

I(f) = Current determined from FIGURE 3

V(f) = Voltage determined from FIGURE 4

B. Variable amplitude so the injected transient can be increased from a minimum to a maximum specified level (that is, at least 10 percent, 50 percent, and 100 percent of the amplitude must be tested at each specified test frequency)

C. Frequency range capable of generating the required test frequencies

D. Current output level of at least 10 A into a 100-ohm load

2. Current probe

3. Oscilloscope with a minimum bandwidth capability of 100 MHz

4. Interconnecting test cable

c. Test setup and procedure. The test setup and procedure shall be as specified in 1 through 3:

1. Functional analysis. A functional analysis shall be performed before the certification testing to identify the items specified in A through C:

A. The interface pin connectors

B. The test sample functions which must be monitored to determine system failure or transient upset



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C. The additional test frequencies (that is, interface cable resonant frequency, clock frequencies, data rates, and so forth)

2. Test setup. The test setup shall be in accordance with FIGURE 6.

3. Test procedures. The test procedures shall be as specified in A through E:

A. Apply the test signal to each interface pin of the test sample, at each test frequency.

B. Monitor the test sample functions in all modes of operation to determine whether there is a failure or transient upset.

C. Inject one transient pulse on each interface pin and at each test frequency.

D. The test sample shall be tested with its power off and on.

E. Record test frequency and level at which a failure or upset occurs.

d. Notes:

1. The test sample shall be tested on a ground plane. There shall be no insulated materials between the test setup and the ground plane.

2. The test cable running between the test sample and the damped sinusoidal generator shall be kept 4 centimeters (cm) to 6 cm above the ground plane.

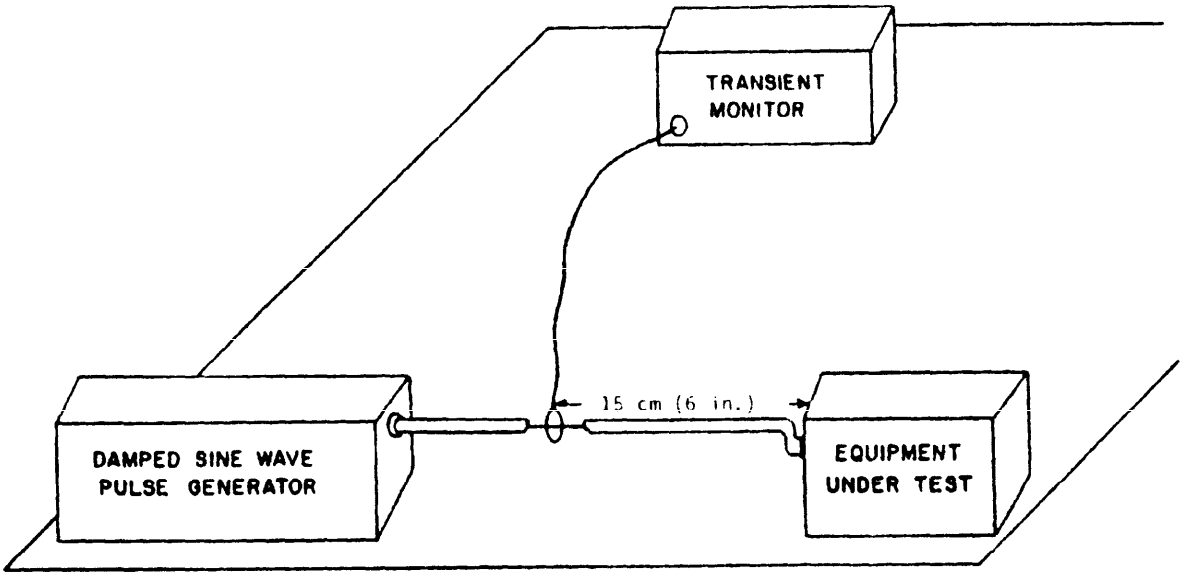
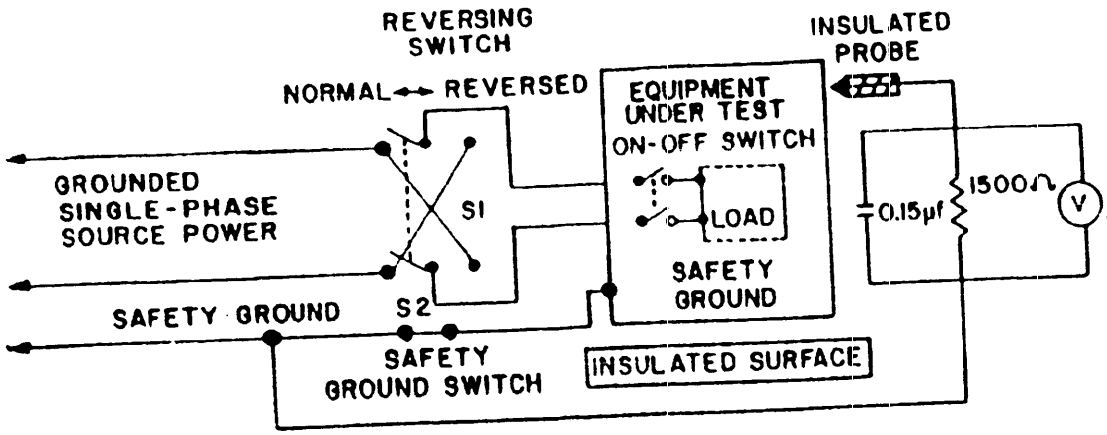



FIGURE 6. EMP test setup.



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**GENERAL ORDER OF TEST:**

1. Source power OFF. Connect equipment per diagram.
2. ON-OFF SW OFF. S1 SW normal. S2 SW closed. Connect source power.
3. OBSERVE WARNING S2 SW OPEN. ON-OFF SW ON.
4. For each probe point, record voltmeter reading (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. Repeat Step 4.
6. S1 SW REVERSED. ON-OFF SW ON. Repeat Step 4.
7. ON-OFF SW OFF. Repeat Step 4.
8. S2 SW CLOSED. S1 SW normal.
9. Repeat Step 3 THRU 8 for each mode of operation.

10. Remove source power. Disconnect equipment. Sign record sheet.

**WARNING**

**DO NOT TOUCH EXPOSED METAL SURFACES**

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

FIGURE 5. Single-phase test diagram for leakage current measurement.





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4.6.11 TEMPEST. Profile TEMPEST tests shall be performed to demonstrate compliance with NACSIM 5100A and NACSIM 5110 requirements. These tests shall consist of at least one each electric field test, magnetic field test, power line conduction test, and signal or control line conduction test.

4.7 Maintainability demonstration. The contractor shall perform a maintainability demonstration at the organizational level. The demonstration shall be performed by qualified technicians, and shall be used to verify conformance to the equipment corrective maintenance requirements.

4.7.1 Demonstration conditions. The maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual printed circuit boards) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. The contractor shall identify 20 candidate corrective maintainability tasks (determined in accordance with APPENDIX A of MIL-STD-471). The information specified in a through c shall be made available for each candidate task:

a. Designation of specific faulty part
b. Failure mode
c. Means of introducing fault (substitution of faulty part or simulation thereof).

The maintainability demonstration shall be conducted under the standard test conditions of MIL-E-16400. The contractor shall use only test equipment, tools, jigs, fixtures, software programs, and technical documentation that are used in the manufacture and checkout of this equipment as specified in the maintainability demonstration plan.

4.7.2 Accept or reject criteria. The accept or reject criteria for the demonstration of the corrective maintenance times shall be as specified in TABLE IV.

TABLE IV. Acceptance criteria.

| | $M_{ct}$ 1/ | $M_{maxct}$ |
|---|---|---|
| Acceptance level | 8 | 0 |
| Sample size | 20 | 20 |

1/Mean-corrective-maintenance-time

Acceptance shall occur when the number of observed corrective maintenance task times, which exceed the required value for each index ($M_{ct}$, $M_{maxct}$), is less than or equal to that specified in TABLE IV, corresponding to each index for the specified confidence level. The duration of each task shall be compared to the required value(s), and determined as greater or less than each index value. The actual value obtained shall also be determined. An acceptance decision can only be made when the values obtained for both $M_{ct}$ and $M_{maxct}$ have been accepted. The accept or reject criteria for preventive maintenance are based on the capability of performing each preventive maintenance task with no degradation in IG performance. Occurrence of IG performance degradation shall be cause for rejection of a preventive maintenance task. The demonstration shall be performed by a technician with experience equivalent to a Navy electronic technician third class (ET-3).

4.8 Workmanship. Workmanship shall be as specified in 4.8.1 through 4.8.2.2.



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4.8.1 General workmanship. The IG, including subassemblies and assemblies, shall be examined for workmanship and soldering during fabrication and assembly for conformance to the requirements of 3.2.4.1 and 3.15.

4.8.2 Workmanship screen. Vibration and temperature cycling shall be performed on each IG as specified in 4.8.2.1 and 4.8.2.2. Vibration shall be performed prior to temperature cycling.

4.8.2.1 Random vibration. The vibration may be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 7. Input vibration levels shall be measured at the mounting points of the item under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density (ASD). The instantaneous acceleration peaks may be limited to three times the rms acceleration level. The item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed.

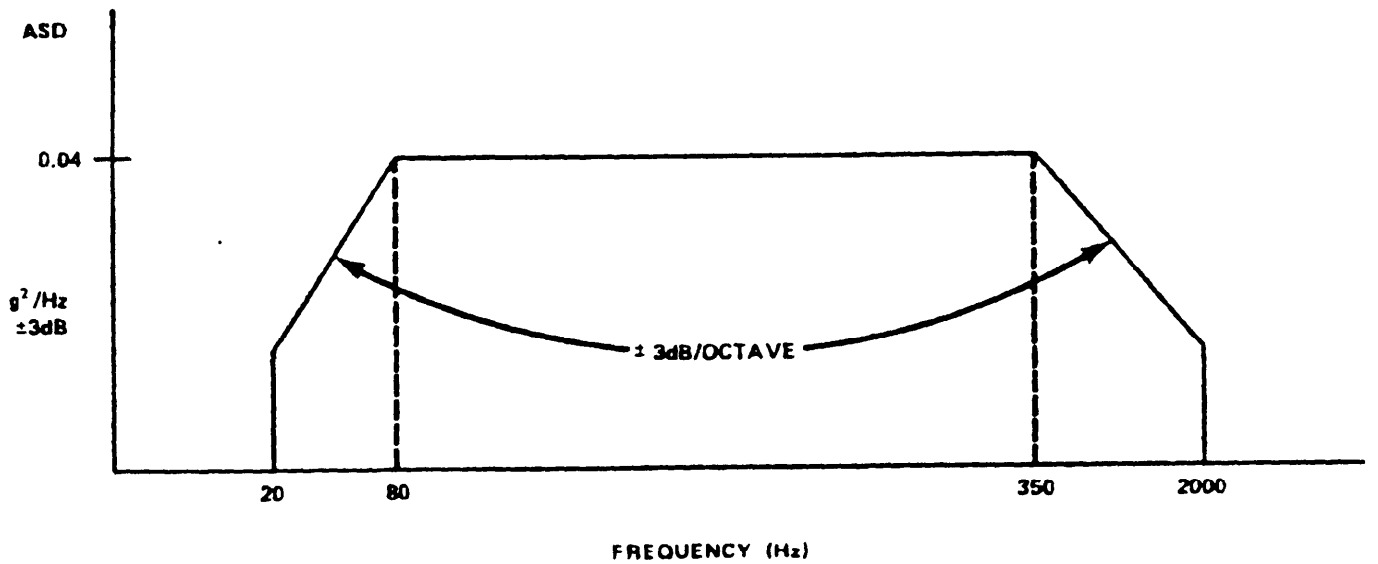



FIGURE 7. Random vibration curve.

4.8.2.2 Temperature cycling. Each IG shall be subjected to 10 cycles of the temperature curve shown in FIGURE 8. The temperature rate of change shall be not less than 5°C per minute. IG power shall be turned on and off at the indicated times. The IG shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the IG reaches 80 percent of the chamber temperature. When failures occur, the IG shall be reworked and the cycling continued for a cumulative total of 10 cycles.



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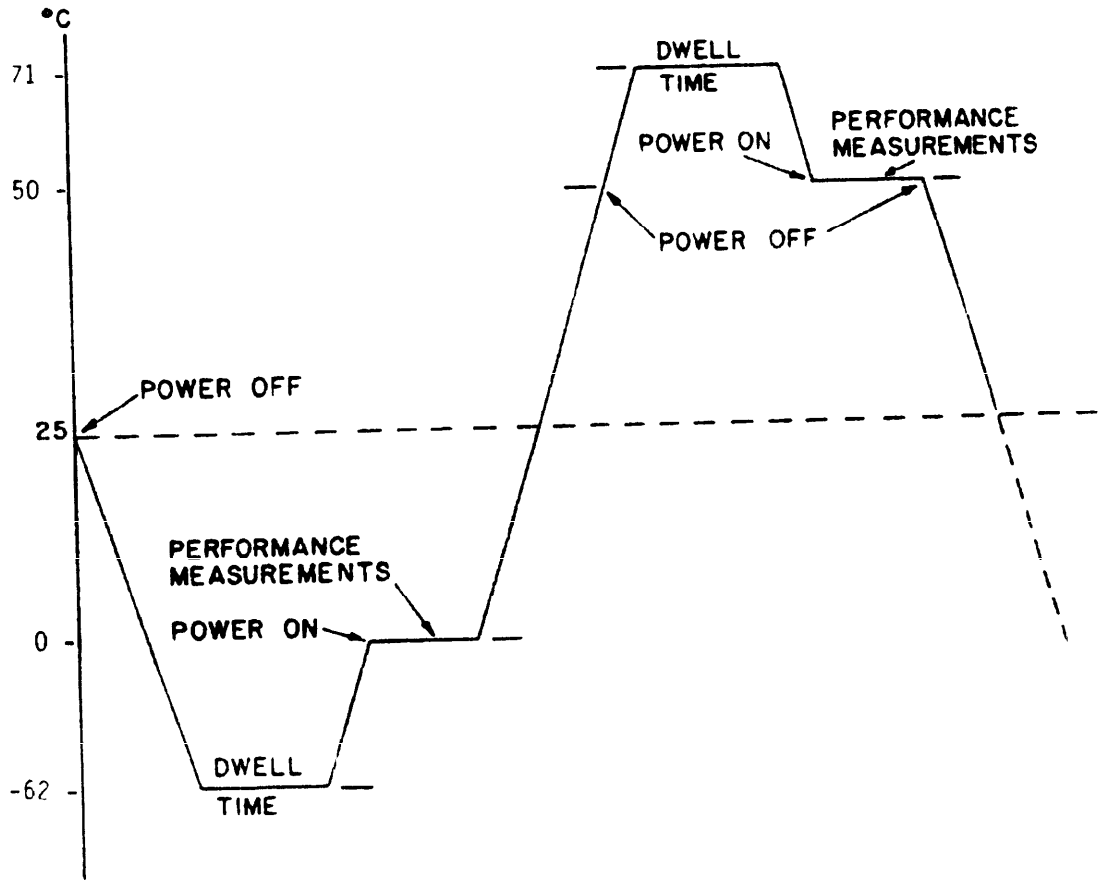



FIGURE 8. One Cycle of temperature curve (not to scale).



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4.8.2.3 Chassis test. After the chassis has been completely fabricated (not including the PC cards) conduct a continuity or short test on all possible combinations of connection points simultaneously, that is, card cages, rear panel connectors, RED/BLACK interface, power supply connections, and so forth.

4.9 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The IG covered by this specification is intended for interconnecting the TTY, crypto, transmitter distributor, typing reperforator, vocoder, tactical data processor, and transceiver. The IG is intended for use on surface and subsurface craft, and at shore sites.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Strapping options required (see 3.3.2.3)
c. Number of first article samples to be submitted (see 4.3)
d. Levels of preservation, packaging, packing, and marking (see 5.1)

6.3 First article. When a first article inspection is required, the item should be a first production item. The first article should consist of the number of units specified in the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.12.

6.5 Special tools. Special tools are defined as those tools not listed in the National Supply Catalog (Copies of this catalog may be consulted in the office of the Defense Contract Administration Service.)

6.6 Subject terms (key word) list.

Altitude
AN/UGC-20 (TTY)
AN/UGC-77, Keyboard Send-Receive (TTY)
AN/WSC-3, Transceiver
Audio, Converter (Vocoder), CV-3333
Built-in test (BIT)
Built-in test equipment (BITE)
Buoy Interface Unit
Crypto, TSEC/KG-36
CV-3333, Audio Converter (Vocoder)
Derating
Distribution statement A
Dust (fine sand)



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Dynamically adaptive receiver transmitters (DARTS)
Electromagnetic compatibility (EMC)
Electromagnetic environmental effects ($E^3$)
Electromagnetic interference (EMI)
Enclosures
Fleet Satellite Communications (FLTSATCOM) Secure Voice
Fungus
Harmonic current
Human engineering
Humidity
Inclination
Interconnecting Group, ON-143(V)6/USQ
Interface RED/BLACK
Keyboard Send-Receive (TTY), AN/UGC-77
Leakage current
Lubricants
Magnetic field
Maintainability
Mean-time-between-failures (MTBF)
Modular construction
Noise, airborne and structureborne
Nonstandard parts
Officer-in-Tactical Command Information Subsystem Exchange (OTCIXS)
ON-143(V)6/USQ, Interconnecting Group
Power and power factor
Power interruption
Prime power design
Random vibration
RED/BLACK Interface
Reliability
Safety
Salt fog
Shock
Spike voltage
Standard parts
Steady-state voltage and frequency
Software
Soldering
Submarine Satellite Information Exchange System (SSIXS)
Tactical Data Processor (TDP)
Temperature
TEMPEST
Thermal Design
Thermal Stress
Tranceiver, AN/WSC-3
Transient frequency
Transient voltage
Transmitter Distributor (TD), TT-187B/UG
TSEC/KG-36 Crypto
TT-187B/UG, Transmitter Distributor (TD)
TT-253/UG, Typing Reperforator
Typing Reperforation, TT-253/UG
Vibration
Welding, structural
Workmanship screen

6.7 Supersession. This specification incluces the requirements of ELEX-I-473 dated 13 April 1981.

Preparing activity:
NAVY - EC

(Project No. 5820-N816)



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